

Breast Cancer Diagnosis Project with ML Methods

1. Problem Definition and Data Collection

Business Problem:

- To develop a machine learning model capable of distinguishing between malignant and benign breast cancer cases with high accuracy.
- To support doctors' diagnostic processes by explaining the model's decisions, providing a more reliable second opinion for patients.
- To increase the effectiveness of treatment processes through early and accurate diagnosis.

Data Source:

- The Wisconsin Breast Cancer (Diagnostic) dataset from Kaggle was used. This dataset is a collection of nuclear features obtained from digitized images of breast masses.
- The dataset was sourced from reliable repositories like the UCI Machine Learning Repository.
- Limitations: The dataset may have been collected from a specific geographic region and time period. This could affect the generalizability of the model across different populations or imaging techniques. The currency of the data should be evaluated in light of new medical developments.

Success Criteria:

- Achieving an F1-Score above a certain threshold, ensuring a high balance between the model's ability to correctly detect malignant tumors (Recall) and its ability to keep the false positive rate low (Precision).
- Ensuring the ROC-AUC value, which indicates the model's power to distinguish between malignant and benign classes, is above a significant limit.
- Finding the model's decisions clinically interpretable and logical through SHAP analysis.

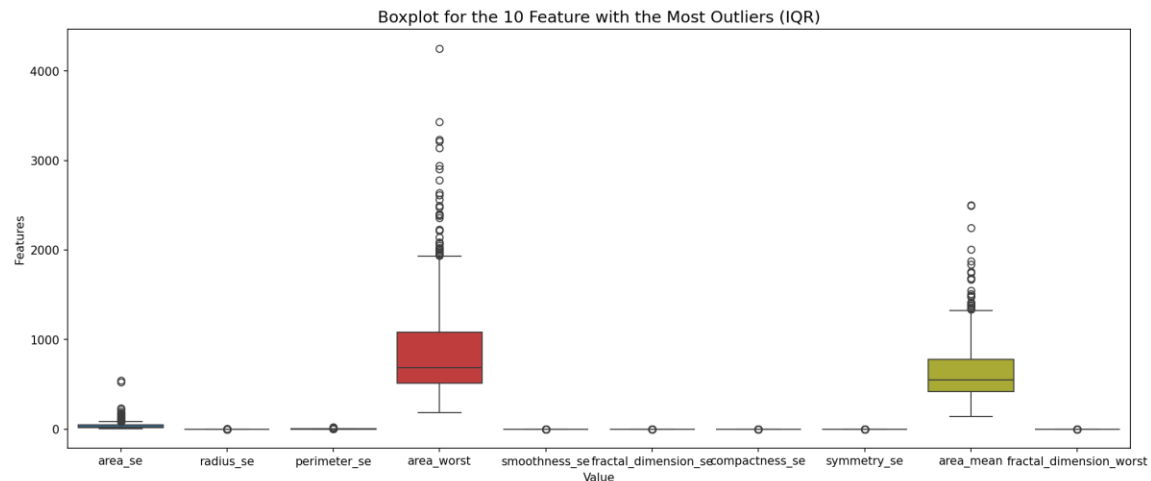
2. Data Preparation and Cleaning

Missing Data Strategy:

- No missing data was found in the dataset; therefore, no Imputer strategy was required.
- However, the dataset contained an irrelevant feature named "Unnamed: 32," which was removed.

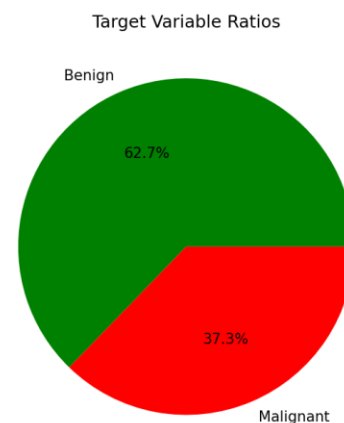
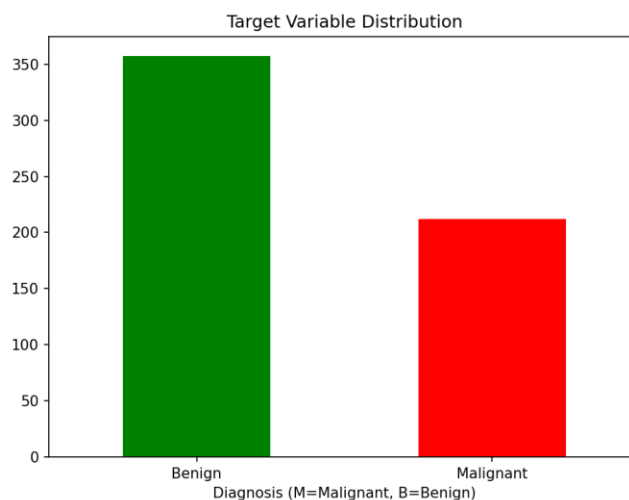
Outlier Management:

- Outliers were identified using the IQR method, and the top ten were visualized.
- In this project, considering the dataset size and the potential information carried by outliers, they were not directly deleted; instead, their effects were minimized through the scaling step (StandardScaler).



Data Transformation:

- Scaling: StandardScaler was utilized.
- Reason: Since features are on different scales (e.g., radius_mean vs fractal_dimension_mean), scaling is crucial for distance-based (e.g., SVM, KNN) and gradient descent-based (e.g., Logistic Regression, Gradient Boosting) algorithms. StandardScaler transforms data to have a mean of 0 and a standard deviation of 1, enabling faster and more accurate learning.
- Categorical Data: There are no categorical features except for the target variable (diagnosis). Using LabelEncoder, the target variable containing 'M' (Malignant) and 'B' (Benign) values was converted into a numerical format (0 and 1) before modeling.

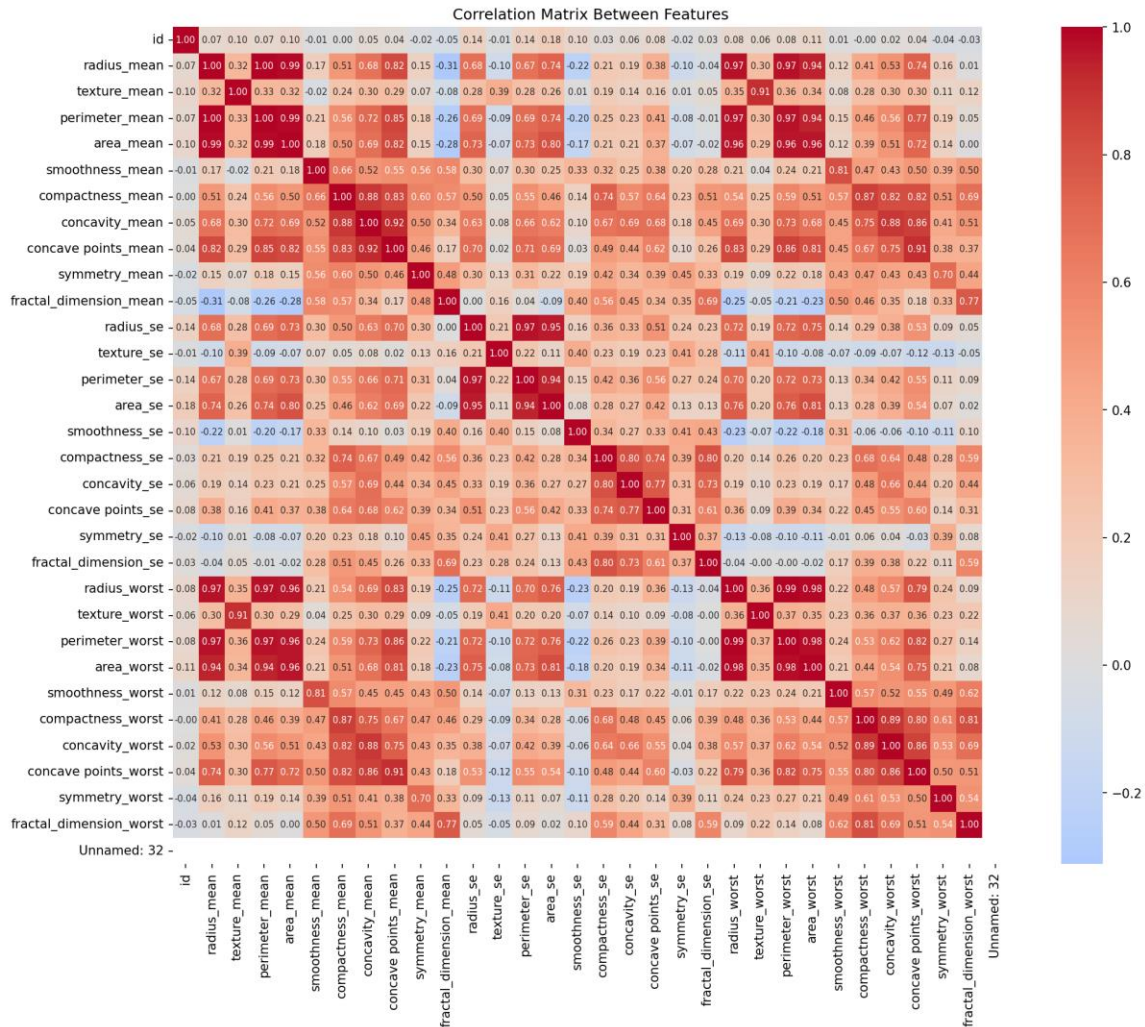


3. Exploratory Data Analysis (EDA) and Feature Engineering

Insights from EDA:

- It was observed that certain features consistently showed higher values in malignant cases.
- For example:

- Malignant tumors exhibited significantly higher values in size-related features such as radius_mean, perimeter_mean, and area_mean compared to benign tumors.
 - Features indicating tumor irregularity and notchiness, such as concavity_mean and concave points_mean, were found to be much higher in malignant cases and showed a strong correlation with the target variable.
 - High correlations exist between some features (e.g., radius_mean and perimeter_mean).
- This suggested a need for caution to prevent overfitting or the consideration of dimensionality reduction techniques like PCA (though not used directly in this project).

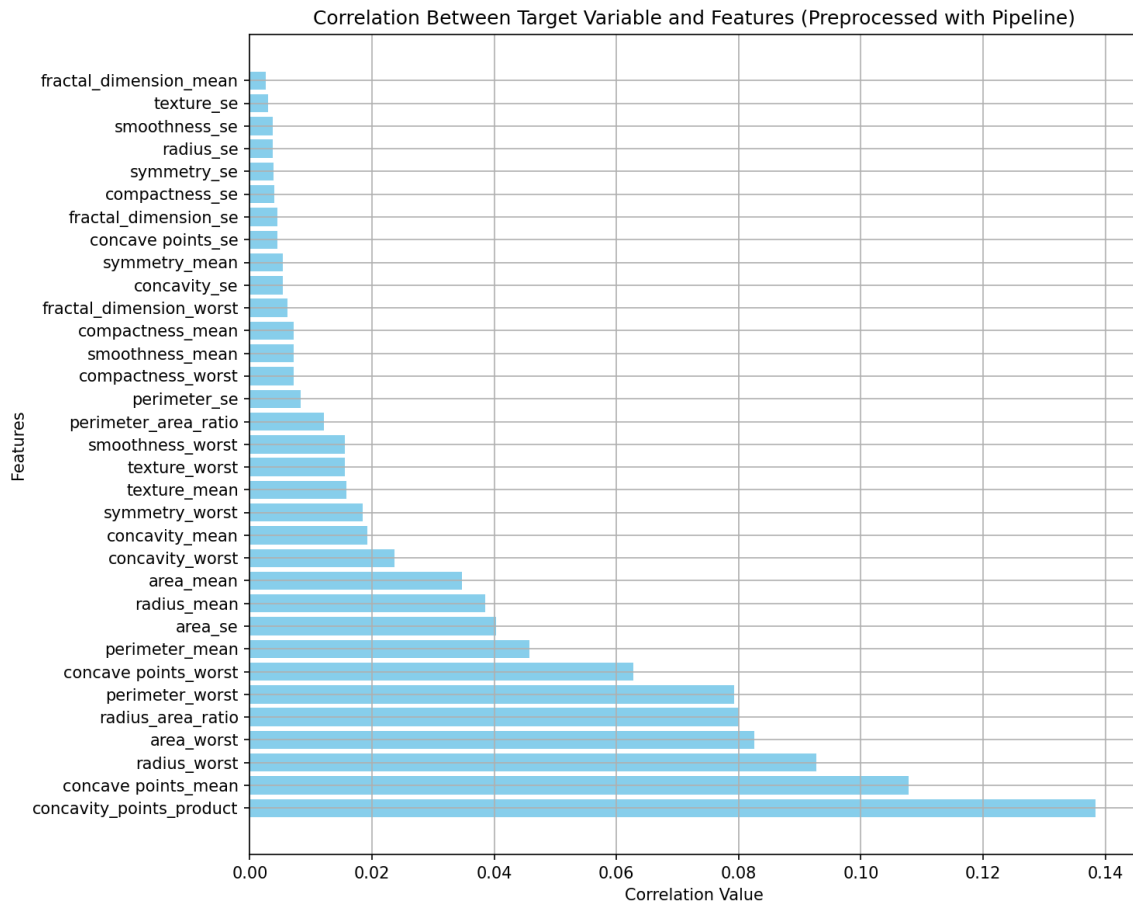


Graph Interpretation:

- The correlation matrix revealed high positive correlations particularly among feature groups with 'mean', 'se', and 'worst' suffixes.
- Feature distribution plots showed that some attributes (e.g., concave points_mean) provide a clear distinction between classes, identifying them as strong predictors for classification.

Feature Engineering:

- New features such as radius_area_ratio, perimeter_area_ratio, and concavity_points_product were derived.
- Reasoning: Since concave points_mean and concave points_worst showed strong relationships with malignancy during EDA, creating concavity_points_product was intended to emphasize the impact of irregularity and sharp edges to help the model learn complex relationships.
- **Contribution:** Correlation analysis via Random Forest showed that these derived features (especially concavity_points_product) ranked among the most important features, providing significant contributions to malignant prediction.



4. Modeling and Evaluation

Metric Selection:

- Why F1-Score and ROC-AUC: In cases like breast cancer diagnosis where class distribution may be imbalanced and both false positives and false negatives have serious consequences, Accuracy alone is insufficient. Therefore, the F1-Score was chosen as the primary success metric. Additionally, ROC-AUC was used to evaluate overall performance across different thresholds.

Model Comparison:

- Nine different classification models were compared: Logistic Regression, SVM, Random Forest, Gradient Boosting, XGBoost, LightGBM, KNN, Naive Bayes, and Decision Tree.
- All models were evaluated on scaled data using F1-Score, Accuracy, Precision, Recall, and ROC-AUC metrics.

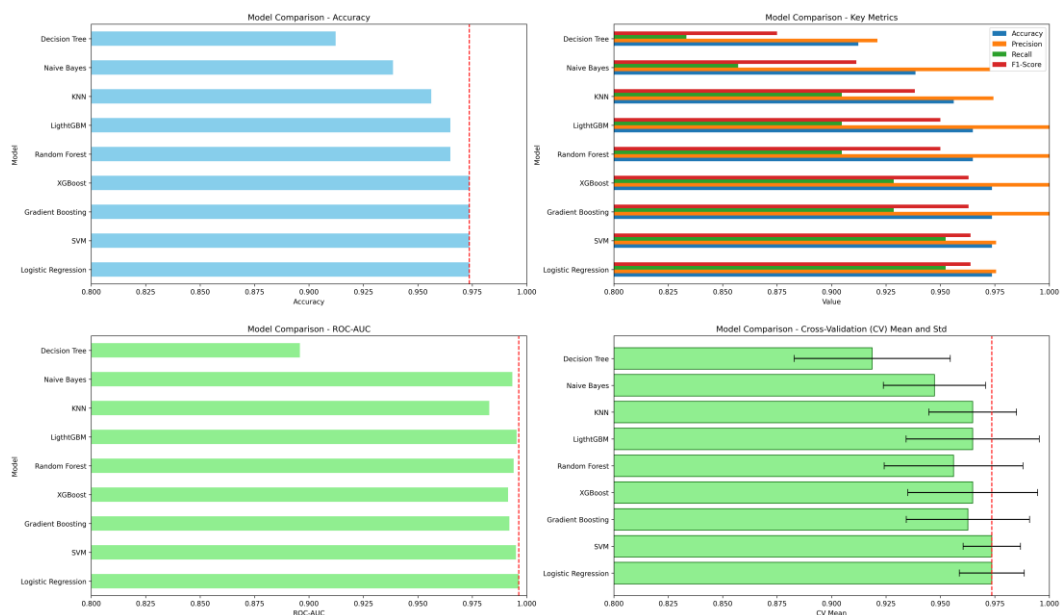
- Reasons for Selecting Logistic Regression:

- **Highest/Balanced F1-Score:** It achieved the best balance between Precision and Recall.
- **High ROC-AUC:** A near-perfect ROC-AUC (approx. 0.996) demonstrated superior discriminative power.
- **Stability:** It showed consistent performance across 5-fold cross-validation with low standard deviation.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	CV_Mean	CV_Std
Logistic Regression	0.973684	0.975610	0.952381	0.963855	0.996362	0.973626	0.014906
SVM	0.973684	0.975610	0.952381	0.963855	0.995040	0.973626	0.013187
Gradient Boosting	0.973684	1.000000	0.928571	0.962963	0.992063	0.962637	0.028317
XGBoost	0.973684	1.000000	0.928571	0.962963	0.991402	0.964835	0.029812
Random Forest	0.964912	1.000000	0.904762	0.950000	0.994048	0.956044	0.031849
LightGBM	0.964912	1.000000	0.904762	0.950000	0.995370	0.964835	0.030612
KNN	0.956140	0.974359	0.904762	0.938272	0.982804	0.964835	0.020143
Naive Bayes	0.938596	0.972973	0.857143	0.911392	0.993386	0.947253	0.023466
Decision Tree	0.912281	0.921053	0.833333	0.875000	0.895833	0.918681	0.035845

Why Other Models Were Not Selected:

- Some tree-based models (e.g., Gradient Boosting, XGBoost) offered very high Precision values, but their Recall values remained slightly below Logistic Regression.
- Models such as KNN and Naive Bayes showed weaker overall performance metrics.
- Ensemble model trials did not significantly improve Logistic Regression's performance and even caused slight decreases in some metrics.



Graph-Based Comparison:

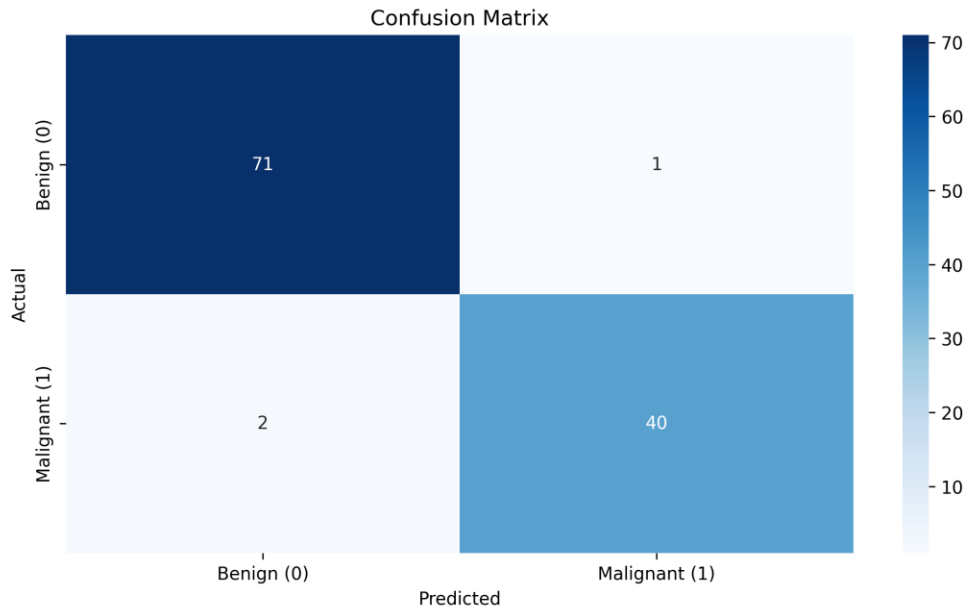
- **Accuracy Graph:** Logistic Regression, SVM, Gradient Boosting, XGBoost, and LightGBM shared first place with an Accuracy of 0.9736.
- **Core Metrics:** Logistic Regression had very close and high Precision and Recall values, meaning it kept false positives low while capturing most true positives.
- In some tree-based models (Random Forest, Gradient Boosting, XGBoost), Precision reached 1.0 but Recall was slightly lower. This means there were very few false positives, but some malignant cases could be missed. Therefore, their F1-Score remained slightly below Logistic Regression.
- **ROC-AUC Graph:** Logistic Regression achieved a ROC-AUC value of 0.9964. Its closest competitors (SVM, Gradient Boosting, Random Forest, and XGBoost) were around 0.99. This indicates that Logistic Regression had near-perfect discriminatory power.
- **Cross-Validation Graph:** Logistic Regression had a CV_Mean of 0.9736 and a low CV_Std, indicating stable and consistent performance across folds. While some models had similar mean scores, their standard deviations were higher, making Logistic Regression stand out in terms of generalizability.

Hyperparameter Optimization:

- GridSearchCV was used for the selected Logistic Regression model. The parameter range for C was [0.001, 0.01, 0.1, 1, 10, 100], penalty options included ['l2'], and solver options included ['lbfgs', 'liblinear'].
- The best parameters were searched using 5-fold cross-validation with F1-Score as the evaluation metric.
- Result: The best parameters were found to be C=10, penalty='l2', and solver='liblinear'. However, the optimized model's performance on the test set (Accuracy: 0.9737, F1-Score: 0.9639, ROC-AUC: 0.9861) was very close to the baseline Logistic Regression model. This indicates that the initial model was already well-tuned and hyperparameter optimization did not provide a meaningful additional gain for this dataset.

Cross-Validation:

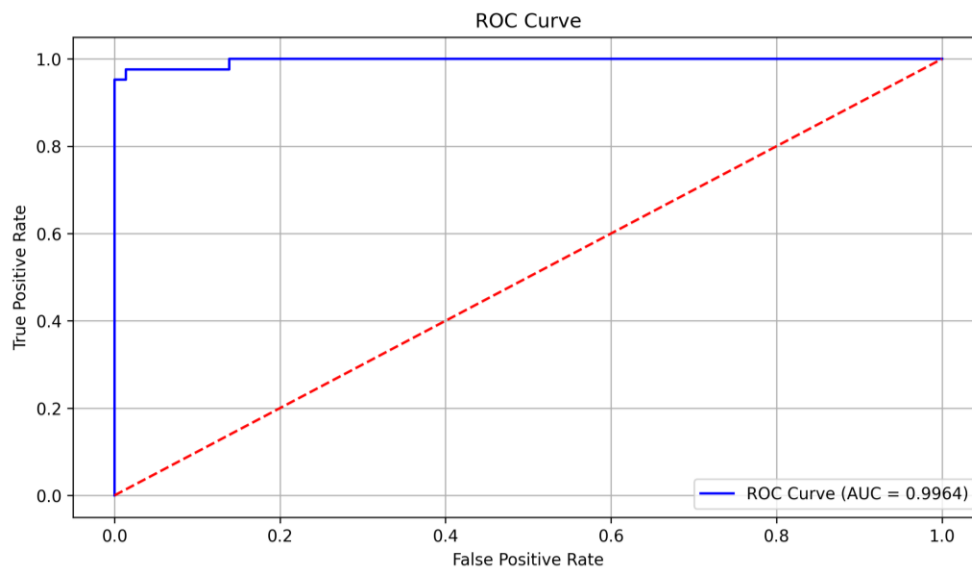
- To ensure that the model was not overfitting and to evaluate its generalizability more robustly, 5-fold Cross-Validation (K-Fold Cross-Validation) was used.
- Performance metrics were calculated in each fold, and both the mean and standard deviation of these metrics were examined.
- The high average performance and low standard deviation of Logistic Regression confirmed that the model performed consistently well across different data subsets and did not show a tendency to overfit.



Detailed Review of Logistic Regression Graphs:

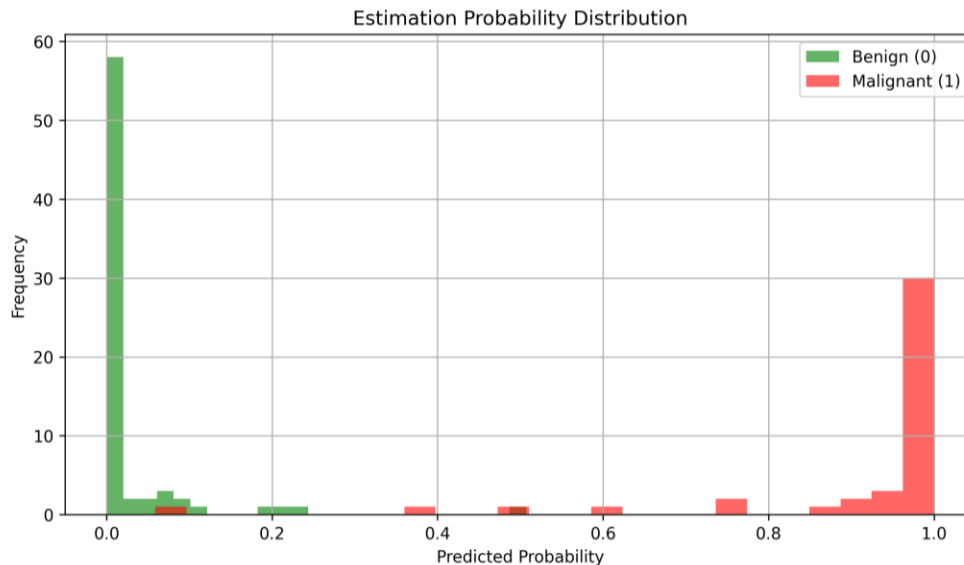
- Confusion Matrix:

- **TN:** 71 benign cases correctly predicted.
- **FP:** 1 benign case incorrectly predicted as malignant (false alarm).
- **FN:** 2 malignant cases incorrectly predicted as benign (critical error).
- **TP:** 40 malignant cases correctly predicted.
- The model labeled only 1 benign case as malignant, but labeled 2 malignant cases as benign, which is the medically riskier part.



- **ROC Curve:** The curve was almost attached to the upper-left corner, with AUC = 0.9964. Even when the threshold changes, the model separates positive and negative classes very

well. This shows that Logistic Regression is suitable for medical decisions where different threshold values may need to be tested.



- **Prediction Probability Distribution:** The class probability distributions were well separated. Benign samples were clustered near 0, while malignant samples were concentrated in the 0.8–1 range. This suggests that the output probabilities are reliable and interpretable.

5. Model Interpretability and Results

Model Interpretation:

- **SHAP (SHapley Additive exPlanations)** analysis was used to explain the decisions of the final Logistic Regression model and increase transparency.

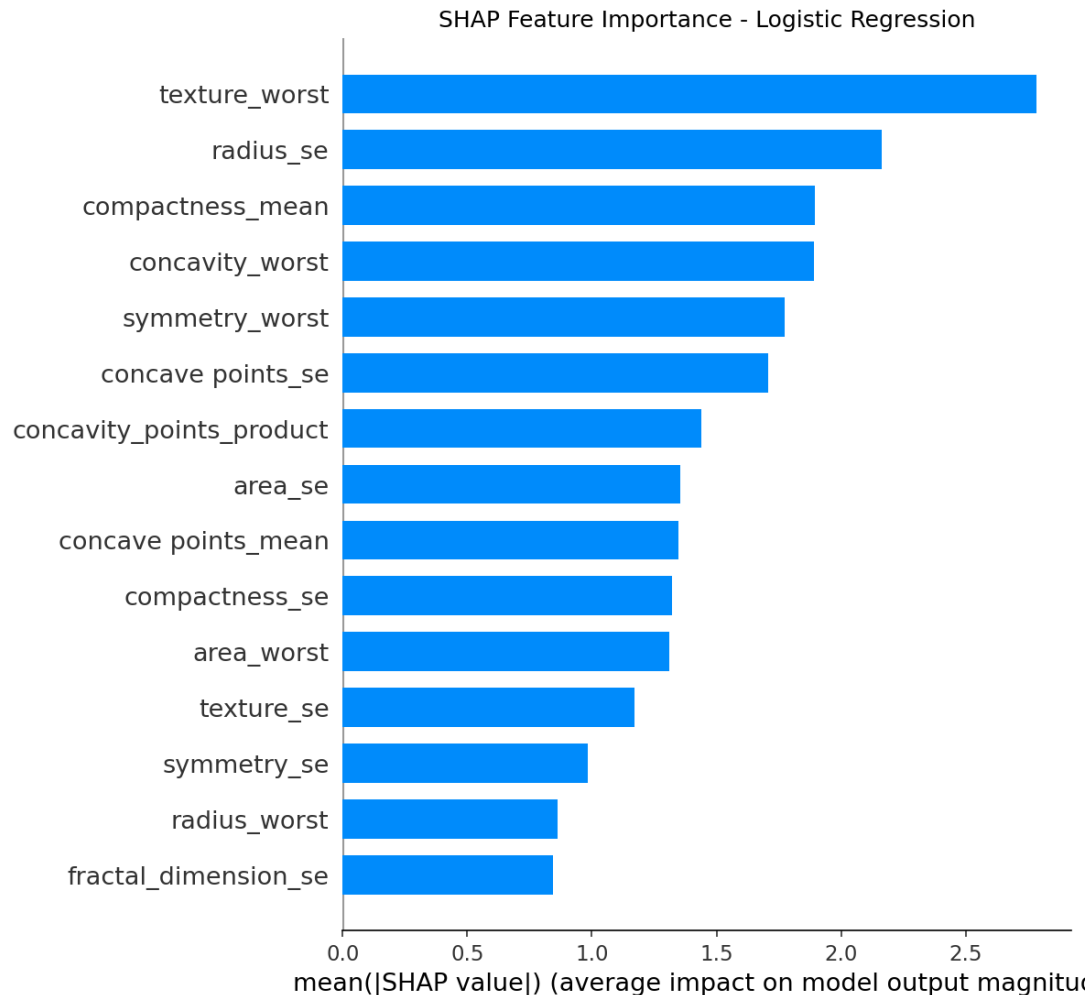
- SHAP enabled quantitative interpretation of each feature's contribution to the model output at both the global level (overall feature importance) and the local level (individual prediction explanations).

Feature Importance (Feature Engineering) – SHAP bar:

- The variables with the greatest impact on the model's decisions include:

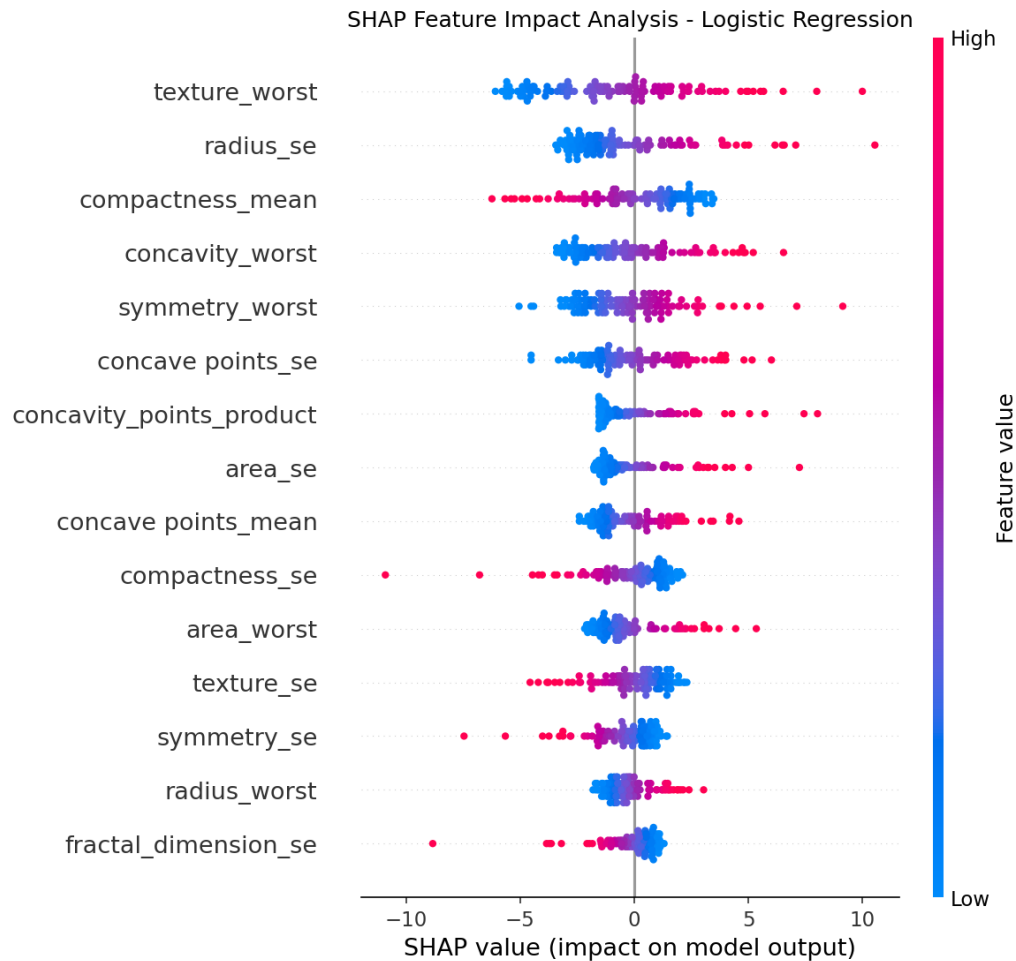
- texture_worst
- radius_se
- compactness_mean
- concavity_worst
- concave_points_se
- concave_points_mean
- concavity_points_product
- symmetry_worst
- area_se

- area_worst
- Such features had the highest average absolute impact on the model's decisions. These findings show that **tumor shape irregularity and size** are the most critical factors in distinguishing malignant from benign cases.



Feature Effect Analysis - SHAP dot:

- This graph can be interpreted as follows:
 - High feature values push the prediction toward the malignant side (positive SHAP).
 - Low feature values push the prediction toward the benign side (negative SHAP).
- For Logistic Regression, features such as texture_worst, radius_se, concavity_worst, concave_points_mean, area_se, and area_worst showed clear patterns:
 - High values (red) mostly appeared on the positive SHAP side, increasing the probability of malignancy.
 - Low values (blue) generally appeared on the negative SHAP side, contributing toward benign predictions.
- Since this graph shows both the ranking of importance and the direction of each feature's effect, it provides a very clear explanation of Logistic Regression's decisions.



Translation of Findings into Business Language:

- The AI model developed in this project offers a powerful support tool for doctors in breast cancer diagnosis.
- Our model can predict whether a tumor is benign or malignant with over 97% accuracy.
- Most importantly, the model not only makes a prediction but can also explain why it made that prediction.
- For example, it can demonstrate that a tumor was marked as malignant due to concrete reasons such as irregularities on the tumor surface or specific size measurements.

What the Model Provides:

- **Second Opinion Tool:** The model can provide doctors with a fast and reliable second opinion, especially in suspicious or borderline cases.
- **Areas of Focus:** The most important findings of the model (such as tumor irregularity and the standard error of the radius) can guide doctors on which features they should focus on more during diagnosis.
- **Educational Use:** Thanks to the model's explainability, it can be used in case analysis training for medical students and newly trained specialists.
- Rather than replacing human expertise, this model aims to ease doctors' workload and

improve the quality of patient care by accelerating the diagnostic process and reducing human error.

6. Project Limitations and Future Work

Weaknesses of the Project:

- The dataset used may have been collected from a specific geographic region and time period. This may limit the model's generalizability across populations with different ethnic backgrounds or images obtained using different medical devices.
- This suggests that the results obtained here are valid only for the current dataset.
- The model relies solely on the features present in the dataset. Additional information used by specialists, such as genetic factors, patient medical history, and lifestyle, was not integrated.

Future Work:

- The model's generalizability can be improved by collecting larger and more diverse datasets from different hospitals and imaging modalities.
- In this study, only numerical data was used. Deep learning-based image analysis models could be integrated.
- By deploying the model into production, a user-friendly interface could be developed.